



DHAAL

Digital Harm Analysis & Alert Layer

A real-time AI shield against digital-arrest & UPI scams — active in the two minutes the scam is actually happening.

ET AI Hackathon 2.0 · Problem Statement 6 — AI for Digital Public Safety

Live demo: dhaal-eta.vercel.app · API: aditya501gupta-dhaal-api.hf.space

THE PROBLEM

A ₹22,495 crore crime wave — every phone is a target

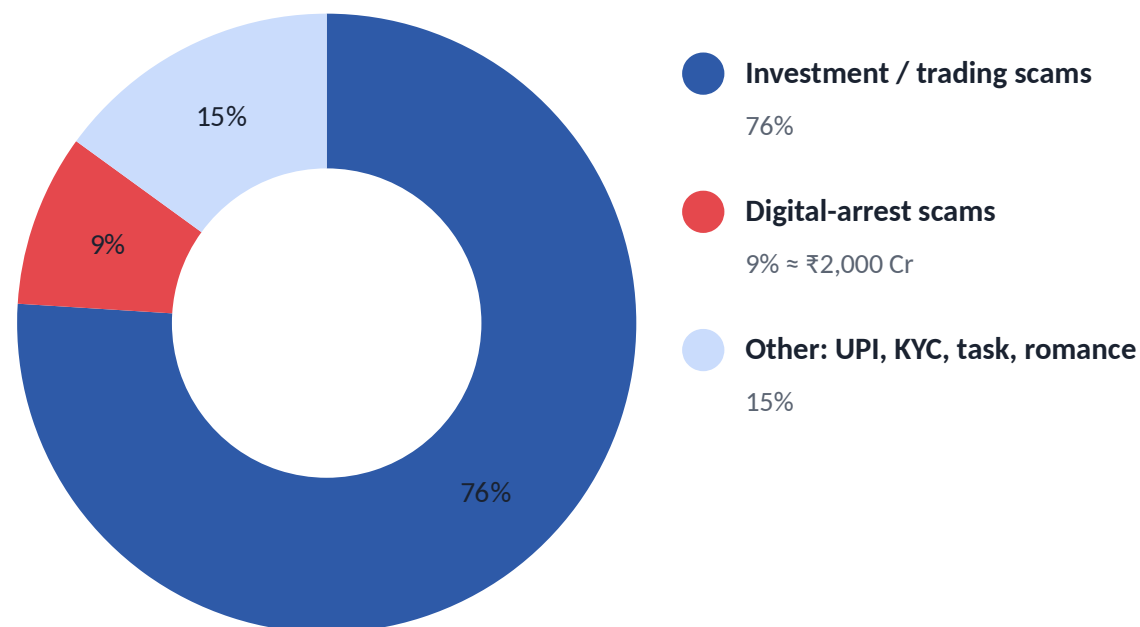
₹22,495

CRORE

lost by Indians to cyber fraud in 2025 — across 28.15 lakh reported cases, a 24% jump in a single year.

Source: I4C / NCRP national data, 2025

Where the money goes (share of 2025 losses)



1 in 5 UPI users has faced fraud — and 51% never report it.

WHY TODAY'S DEFENSES FAIL

By the time a bank or the police can act, the money is gone

1

The window is ~2 minutes

The scam is a live performance — authority, fear, urgency, secrecy — engineered to make a person act before they can think.

2

The victim is isolated

“Don't tell anyone, stay on the call.” The one moment a second opinion would help is the exact moment it's designed away.

3

Help is post-mortem

Blocklists, 1930 and bank freezes all act after the transfer. Nothing intervenes in the citizen's hand, in their language, in the moment.

DHAAL closes exactly that gap — a second opinion in the seconds that matter.

One shield, three layers



Citizen Shield

Paste, share or screenshot any message → an explained verdict in seconds: scam or not, the manipulation tactics, official advice, and a ready-to-file 1930 complaint. 6+ languages.



Intelligence

PII-redacted reports cluster into fraud campaigns via script embeddings + HMAC-pseudonymised indicators — early warning after as few as 5 reports.



Command Centre

Geospatial hotspots, campaign explorer and SHA-256 hash-chained evidence dossiers with BSA s.63-style certificates for investigators.

Live today: Layer 1 is deployed and running. Layers 2–3 are the build-out.

LIVE TODAY — NOT A MOCKUP

Open it on your phone right now



Verdict in under 3 seconds

Paste any SMS / WhatsApp / email — get SCAM • SUSPICIOUS • SAFE with a confidence score.



Scam anatomy, in their words

The manipulative phrases are highlighted inside the message, with the tactics named.



English ⇌ हिंदी

Whole interface, samples and warnings switch language in one tap.



One-tap 1930 complaint

Auto-drafts a [cybercrime.gov.in](https://www.cybercrime.gov.in) / 1930-ready complaint, and a share-warning for family.



DHAAL

"This is FedEx. Your parcel has illegal items.
Mumbai Police issued an arrest warrant..."



SCAM

digital arrest • 98%

authority impersonation

fear + urgency

isolation

payment pressure

Draft 1930 complaint

Try it: dhaal-eta.vercel.app • API + live docs: aditya501gupta-dhaal-api.hf.space/docs

DESIGNED FOR THE PEOPLE SCAMMERS TARGET MOST

Consent-first, family-aware, and built for elders



Consent first

A 6-language onboarding screen with an explicit “what DHAAL will never do” promise. Every scan is opt-in and off by default.



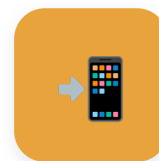
Family guardian

Set a trusted relative. On a SCAM verdict the phone vibrates and offers a one-tap WhatsApp / call warning to them.



Easy-view mode

One switch enlarges text and buttons for elderly users — the people digital-arrest gangs target most.



Install as an app

Add-to-home-screen PWA — opens like a native app, no store needed, works on any phone.

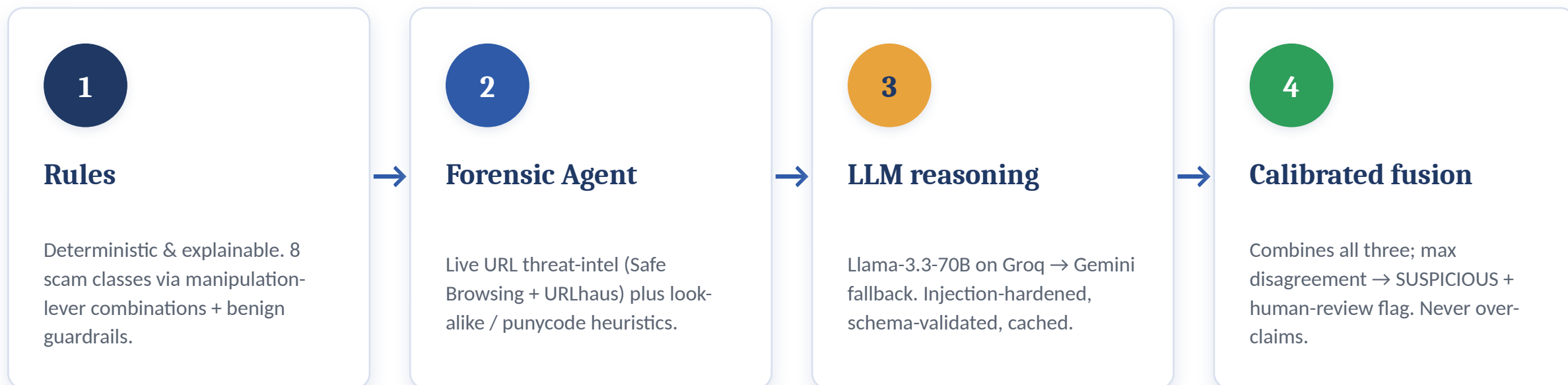


Native SMS shield

The Android build scans incoming SMS in the background and fires a vibrate + alert even when the app is closed.

Privacy: message and call text is analysed over HTTPS and never stored on the device — consent is built in, not bolted on.

How a verdict is built — four signals, honestly fused



SSRF-safe by design: the Forensic Agent never fetches or opens the scammer's link — it only asks trusted threat databases about it, and strips any victim PII from the URL before it ever leaves the box.

DOES IT ACTUALLY WORK?

IndiaScam-Bench — measured, and reproduced on every commit

100%

scam recall

100%

precision

0%

false-positive rate

103

labelled samples
73 scam / 30 benign
EN · HI · Hinglish

Honest caveat: the seed corpus is small and partly synthetic. It grows toward 600+ samples with a temporal hold-out split before any headline generalisation claim — and the whole benchmark re-runs in CI on every push, so these numbers can't silently rot.

Detects 8 scam classes: digital arrest · KYC/re-verify · UPI collect · task/investment · loan · lottery · sextortion · delivery-OTP.

TRUST IS THE PRODUCT

Safe & private by design



No SSRF hole

Treats every link as an untrusted string — never fetches or resolves it. Private/loopback hosts are flagged, never contacted.



PII never leaves the box

Query strings carrying phone numbers, OTPs and tokens are stripped before any third-party lookup.



Prompt-injection hardened

Message content is data, never instructions — the LLM can't be hijacked by text inside the scam.



DPDP-mapped & pseudonymised

Consent-based, auto-redaction before storage, HMAC-only indicators — no raw scammer identifiers in the graph.

A misbehaving or offline feed degrades gracefully to offline heuristics — it can never crash a verdict.

Built to reach every phone — at near-zero marginal cost

₹0

infrastructure cost

Groq, Gemini, Vercel & Hugging Face free tiers
— the rules engine is dependency-free.

6+

languages

Explainable output in the citizen's own language
— designed for Bharat, not just metros.

Any

channel

SMS, WhatsApp, email, screenshots and (next)
live-call audio — one verdict engine behind all
of them.

The prize isn't detection accuracy alone — it's putting an explainable second opinion in the hand of the person on the call, before the transfer. That is a public-safety utility, not a product.

WHAT'S NEXT

From a working shield to a national early-warning system



NOW

L1 shield + native app

6-language app, consent, guardian alerts, elderly mode, hybrid API and a native Android SMS shield — all live, CI-green.



NEXT

L2 campaign clustering

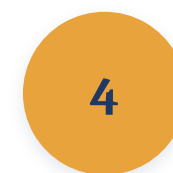
Script-embedding clusters raise an early warning after ≤ 5 reports.



THEN

L3 command centre

Geo hotspots + SHA-256 hash-chained evidence dossiers for police.



SOON

More languages + on-device

Bengali/Tamil/Telugu/Marathi corpus, and on-device live-call alerts.

Already done: IndiaScam-Bench grown to 673 samples with a temporal hold-out; 100% recall on the real hold-out, and survived a 32-case red-team.



A shield in the two minutes that actually matter.

Rules + Forensic Agent + LLM, honestly fused — explainable, private, and live today.

CITIZEN APP

dhaal-eta.vercel.app

LIVE API + DOCS

aditya501gupta-dhaal-api.hf.space/docs

CODE

github.com/aditya501gupta/dhaal